

Joonho Lee

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WORK

Neuromeka AI group ai.neuromeka.com

April 2024 - Now

Group Leader

RIVR rivr.ai

Jan. 2024 - Now

Advisor for RL-based control and navigation

EDUCATION

Robotic Systems Lab, ETH Zurich

Nov. 2018 - Feb. 2024

PhD in Mechanical Eng.

ETH Zurich

Feb. 2017 - Sept. 2018

Master of Science in Robotics, Systems and Control (Rank: 1/76)

Pohang institute of Science and Technology (POSTECH)

Mar. 2012 - Aug. 2016

Bachelor of Science, Department of Mechanical Engineering (Rank: 2/17)

Magna Cum Laude

SELECTED PAPERS

[Google Scholar Profile](#)

1. **Joonho Lee**, Yunho Kim, Seokjoon Kim, Quan Nguyen, and Youngjin Heo. Learning fast, tool-aware collision avoidance for collaborative robots. *IEEE Robotics and Automation Letters*, 2025
2. Sehoon Ha, **Joonho Lee**, Michiel van de Panne, Zhaoming Xie, Wenhao Yu, and Majid Khadiv. Learning-based legged locomotion: State of the art and future perspectives. *The International Journal of Robotics Research*, 44(8):1396–1427, 2025
3. Takahiro Miki, **Joonho Lee**, Lorenz Wellhausen, and Marco Hutter. Learning to walk in confined spaces using 3d representation. In *2024 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2024

IEEE ICRA Best Paper Award in Service Robotics

4. **Joonho Lee***, Tianxu An*, Marko Bjelonic, Flavio De Vincenti, and Marco Hutter. Scalable multi-robot cooperation for multi-goal tasks using reinforcement learning. *IEEE Robotics and Automation Letters*, 2024
**Equal contribution*
5. **Joonho Lee***, Lukas Schroth*, Victor Klemm, Marko Bjelonic, Alexander Reske, and Marco Hutter. Exploring constrained reinforcement learning algorithms for quadrupedal locomotion. In *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 11132–11138. IEEE, 2024
**Equal contribution*
6. **Joonho Lee**, Marko Bjelonic, Alexander Reske, Lorenz Wellhausen, Takahiro Miki, and Marco Hutter. Learning robust autonomous navigation and locomotion for wheeled-legged robots. *Science Robotics*, 9(89):eadi9641, 2024
7. **Joonho Lee***, Alvaro Belmonte-Baeza*, Giorgio Valsecchi, and Marco Hutter. Meta reinforcement learning for optimal design of legged robots. *IEEE Robotics and Automation Letters*, 2022
**Equal contribution*

8. Takahiro Miki, **Joonho Lee**, Jemin Hwangbo, Lorenz Wellhausen, Vladlen Koltun, and Marco Hutter. Learning robust perceptive locomotion for quadrupedal robots in the wild. *Science Robotics*, 7(62):eabk2822, 2022

Project page: leggedrobotics.github.io/rl-perceptiveloco/

9. **Joonho Lee**, Jemin Hwangbo, Lorenz Wellhausen, Vladlen Koltun, and Marco Hutter. Learning quadrupedal locomotion over challenging terrain. *Science robotics*, 5(47), 2020

Project page: leggedrobotics.github.io/rl-blindloco/

IEEE RAS 2020 TC on Model-Based Optimization for Robotics Best Paper Award

10. Vassilios Tsounis, Mitja Alge, **Joonho Lee**, Farbod Farshidian, and Marco Hutter. Deepgait: Planning and control of quadrupedal gaits using deep reinforcement learning. *IEEE Robotics and Automation Letters*, 5(2):3699–3706, 2020
11. **Joonho Lee**, Jemin Hwangbo, and Marco Hutter. Robust recovery controller for a quadrupedal robot using deep reinforcement learning. *arXiv preprint arXiv:1901.07517*, 2019
12. Jemin Hwangbo, **Joonho Lee**, Alexey Dosovitskiy, Dario Bellicoso, Vassilios Tsounis, Vladlen Koltun, and Marco Hutter. Learning agile and dynamic motor skills for legged robots. *Science Robotics*, 4(26):eaau5872, 2019

HONORS

1. ETH Zurich Medal 2025 for outstanding Doctoral Thesis in ETH Zurich, 2025.
2. IEEE ICRA Best Paper Award in Service Robotics Best Paper Award, 2024.
3. IEEE Robotics & Automation Society 2020 TC on Model-Based Optimization for Robotics Best Paper Award.
4. ETH Zurich Medal 2019 for outstanding Master’s Thesis in ETH Zurich, 2019.
5. Willi Studer Prize 2019 (Best Graduate Award) in Robotics, Systems and Control, D-MAVT in ETH Zurich, 2019.
6. Vice President of the 28th Student Council of POSTECH, 2014
7. Korean National Scholarship for Science and Engineering, 2012–2015